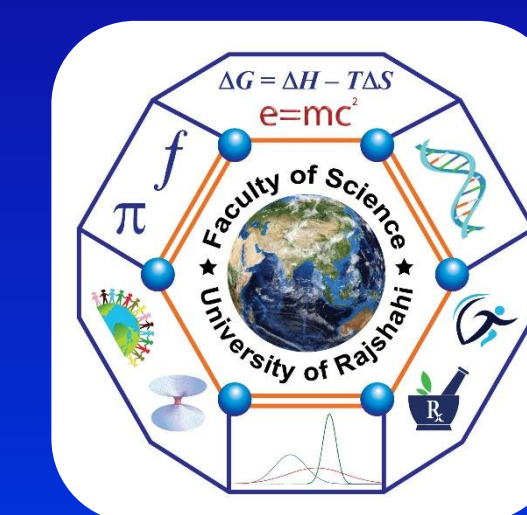




Isolation and Characterization of Rhizospheric and Endophytic Bacteria from Maize Plants for Potential Usage in Sustainable Agriculture



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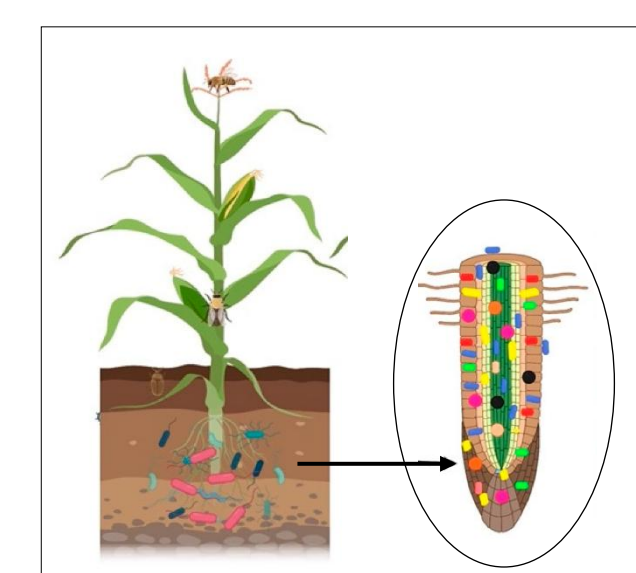
ABSTRACT

The population of the world is increasing and to feed this rapidly growing population, we need to boost agricultural productivity through sustainable, eco-friendly approaches. Such approaches include using nutrient-mobilizing bacteria to reduce chemical fertilizer dependence and antagonistic bacteria as biocontrol agents against pathogens. Maize (*Zea mays*) is the most multi-purpose and agro-climatic situation-adapted crop cultivated in approximately 160 countries. For the desired yield, maize cultivation needs nitrogen, potassium, phosphorus, and compound fertilizer. These fertilizers can actively participate in soil and water contamination which is highly hazardous for human beings and the environment. The rhizospheric and endophytic bacteria from maize have been used as potential plant-growth-promoting and biocontrol agents for sustainable agriculture in several countries. The primary goal of this study was to isolate and characterize the rhizospheric and endophytic bacteria from maize plants and evaluate the efficacy of the isolated bacteria in plant growth promotion. A total of thirteen bacterial isolates were obtained from maize plants grown in Bangladesh and differentiated through several morphological and biochemical analyses. The growth-promoting activities of the bacteria were evaluated through eight in vitro plant growth-promoting screening assays. Four isolates showed better results and were therefore selected for in vivo assay. Those isolates improved plant growth-promoting parameters including germination rate, germination index, vigor index, root and shoot height and weight, chlorophyll and carotenoid content, water content, and deficit and reactive oxygen species (ROS) scavenging activity. These findings highlight the potential of these bacteria for agricultural applications, particularly in promoting the growth and yield of maize plants.

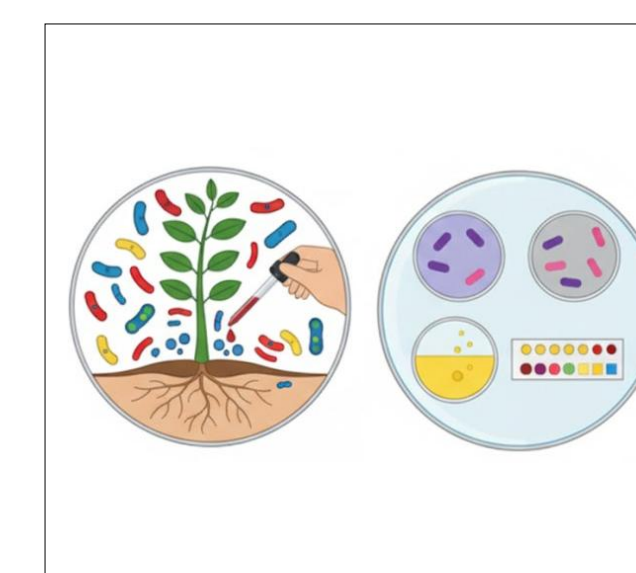
INTRODUCTION

Maize is an important crop worldwide for food, livestock feed and several industrial applications. It is believed to have originated about 10,000 years ago and is the third largest grown cereal crop after wheat and rice. Carbohydrates, proteins, fats, vitamins and minerals are the major constituents of maize (1) and also contains bioactive compounds to fight against chronic diseases. US and China are the predominant producers with more than 160 countries growing this crop. Maize is broadly adopted in Bangladesh and has a greater yield compared to the other cereals (6.15 t/ha) (2). It grows best in loam soil with neutral pH and high organic content. But, over application of chemical fertilizer (N, P and K) is equal with maize production leading to soil and water pollution, greenhouse gas emissions and poor sustainability. Therefore, PGPR are an environmentally friendly choice. These good bacteria organisms such as *Bacillus*, *Pseudomonas* and *Rhizobium* help promote growth of plants, nutrient uptake and disease resistance (3). Maize is susceptible to a variety of diseases such as gray leaf spot, anthracnose and smut. The present study aims to isolate and characterize rhizospheric and endophytic maize bacteria from Bangladesh for their possible utility in sustainable agriculture and growth promotion.

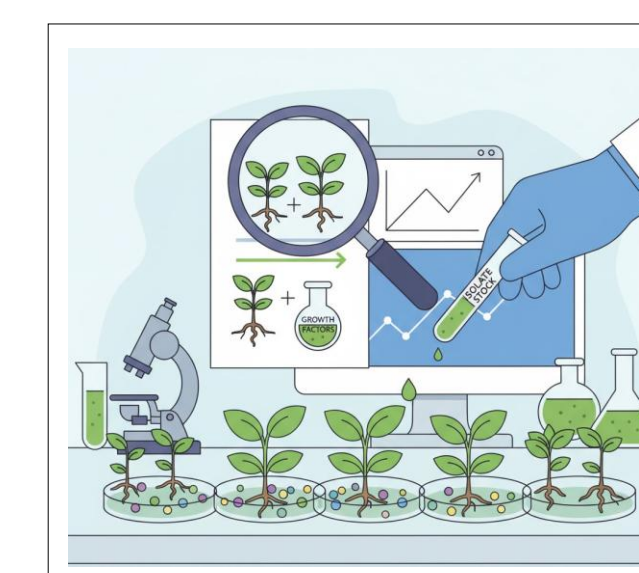
OBJECTIVES



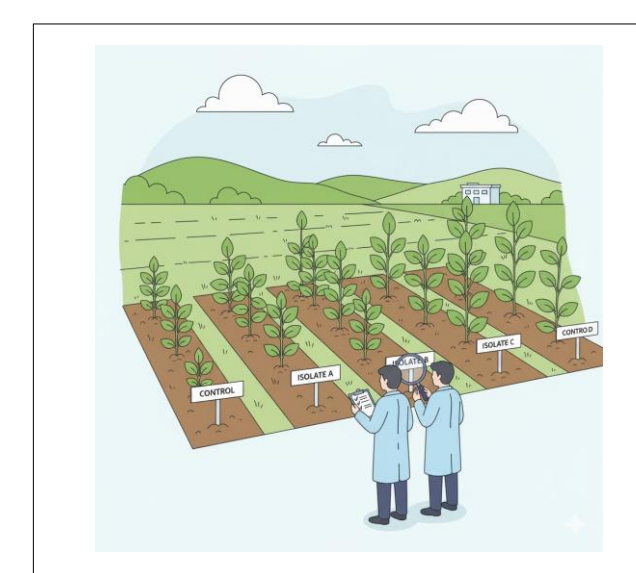
To isolate the rhizospheric and endophytic bacteria from maize



To characterize rhizospheric and endophytic bacteria through morphological and biochemical tests

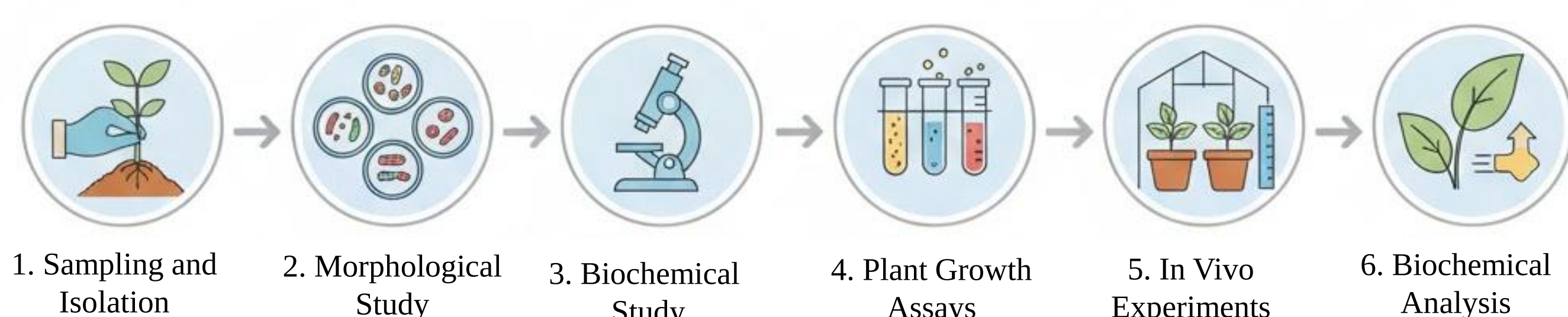


To screen the isolates for plant-growth-promoting qualities



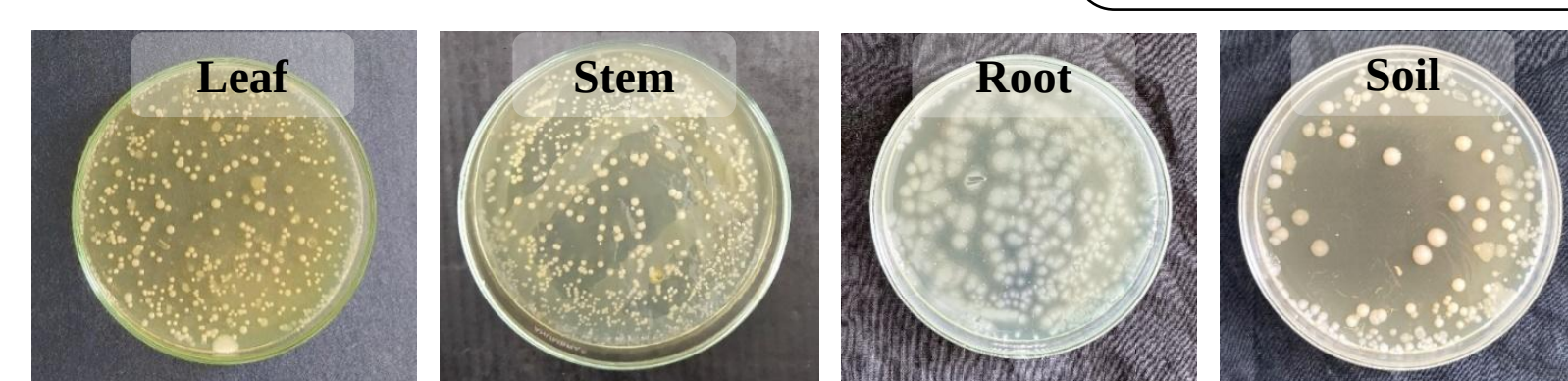
To evaluate the efficacy of selected isolates in plant growth promotion

METHOD

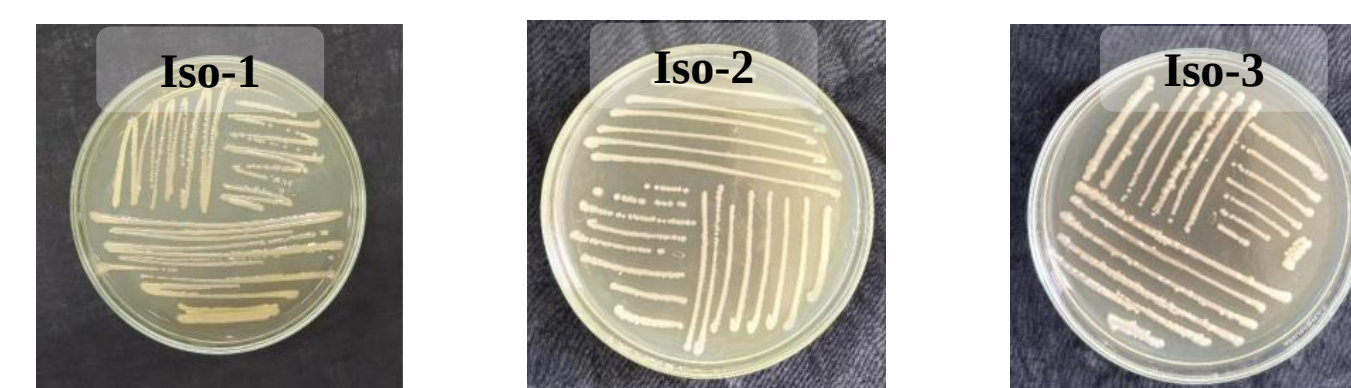


RESULTS and DISCUSSION

Isolation of Bacteria

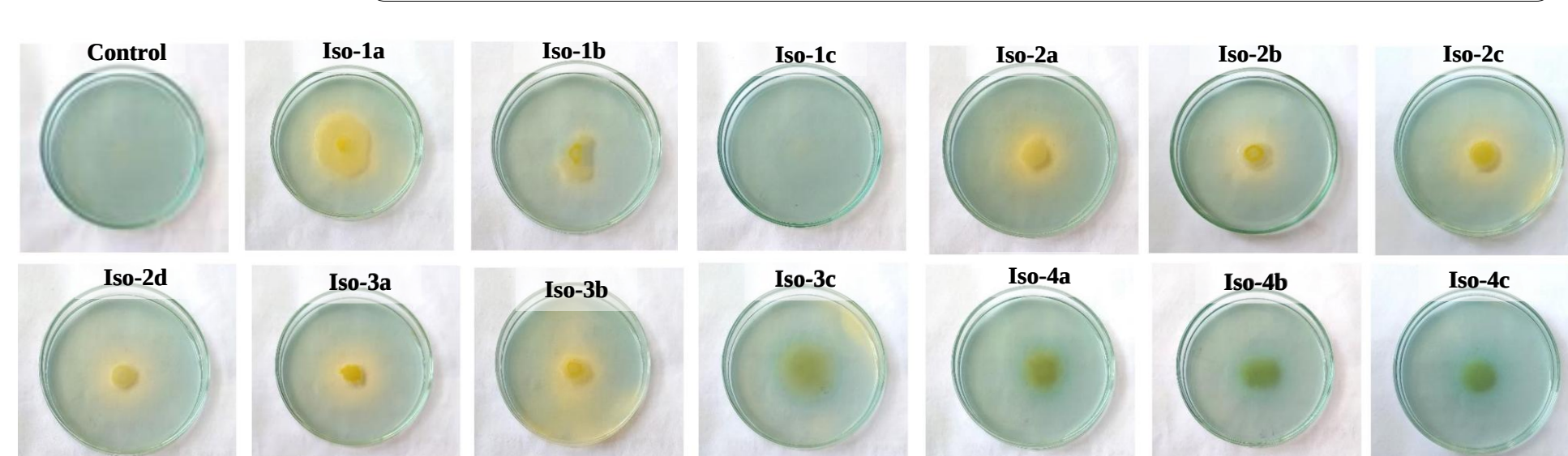


Spreading Plates

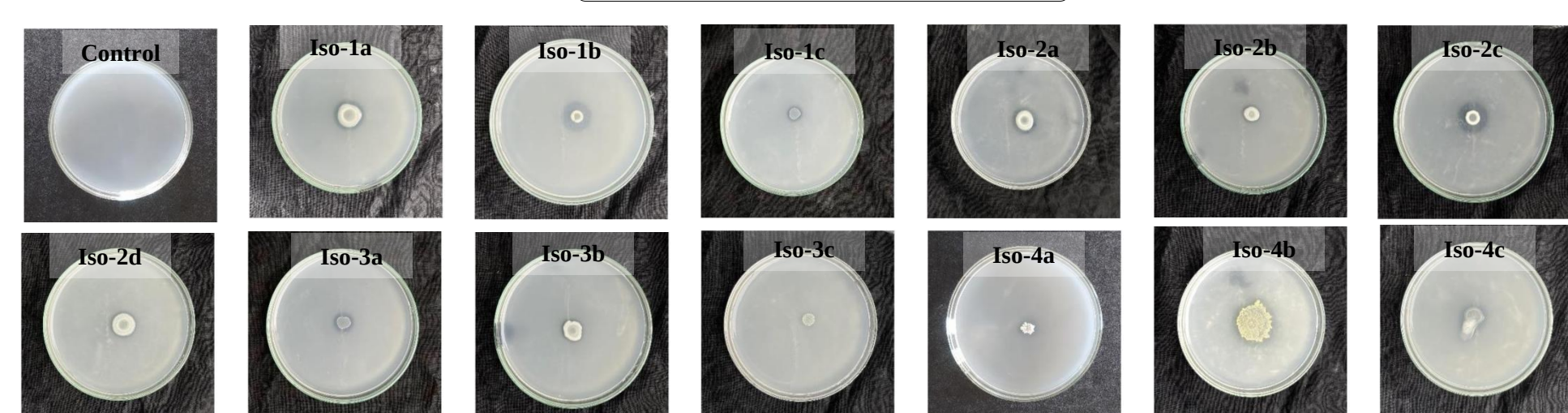


Streaking Plates

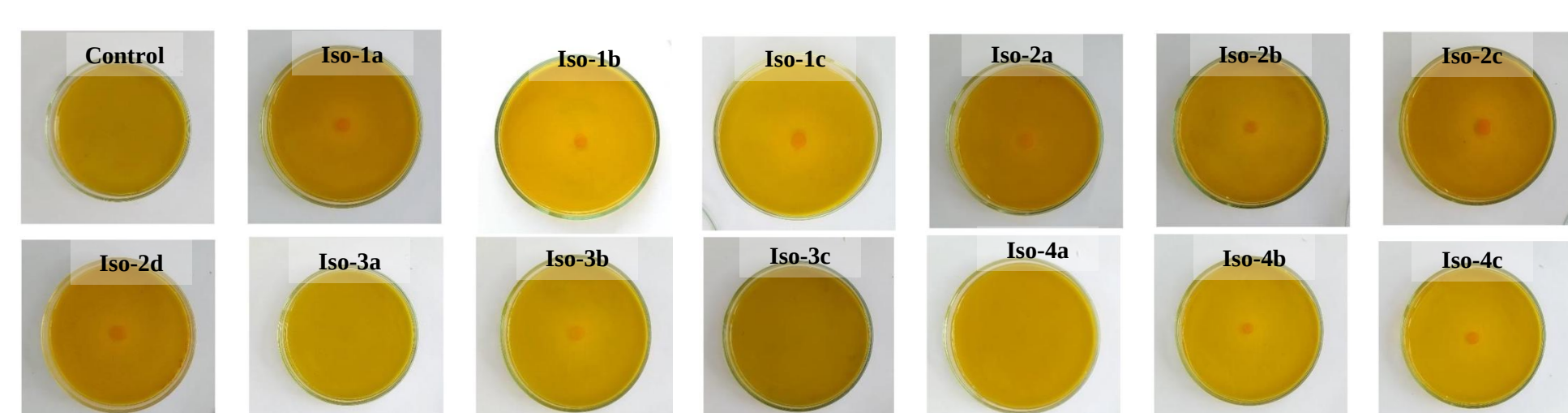
Plant Growth Promoting Traits of Isolated Bacteria



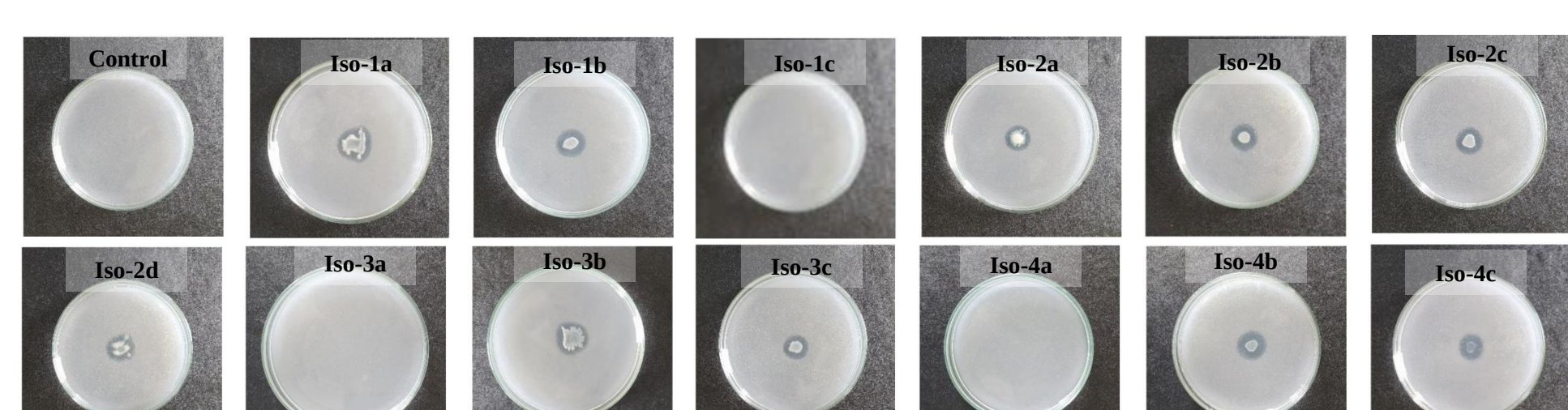
Nitrogen Fixation Test



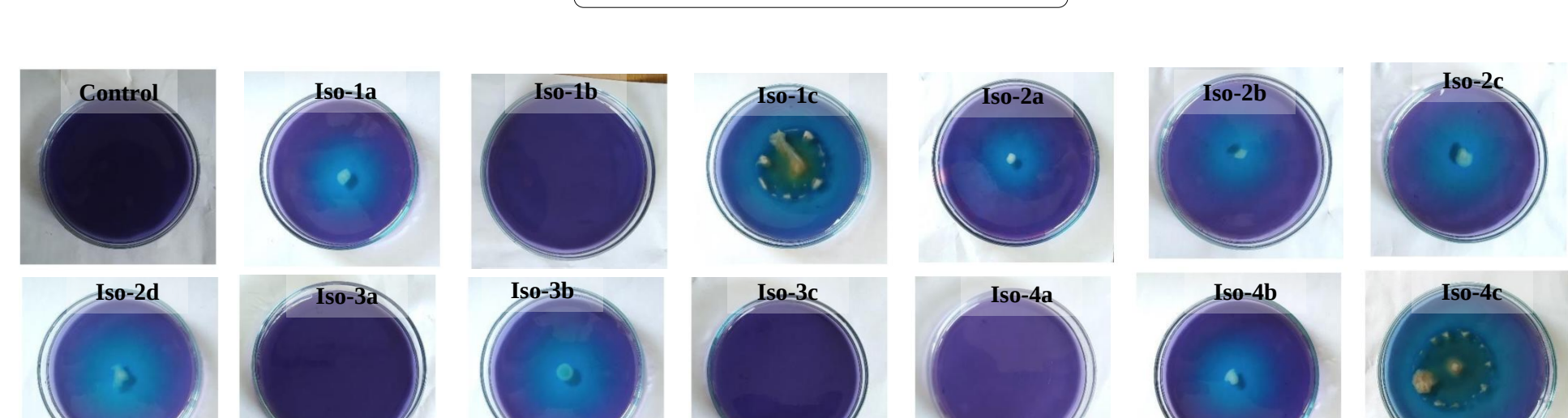
Phosphate Solubilization Test



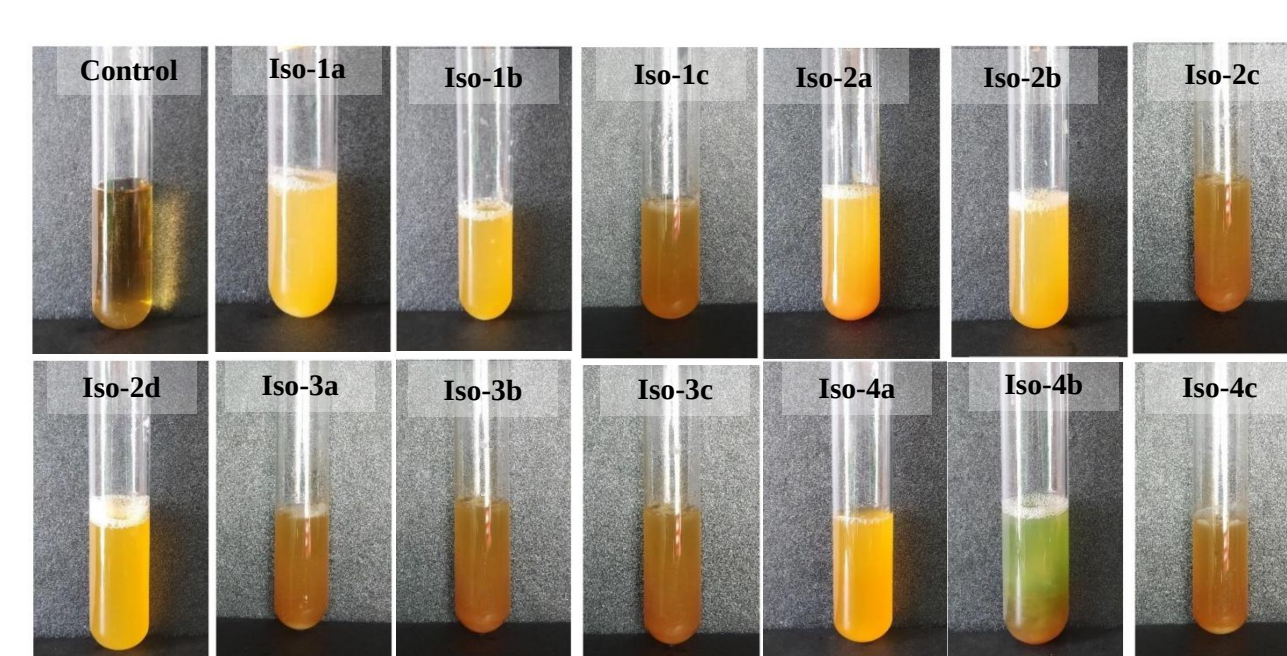
Potassium Solubilization Test



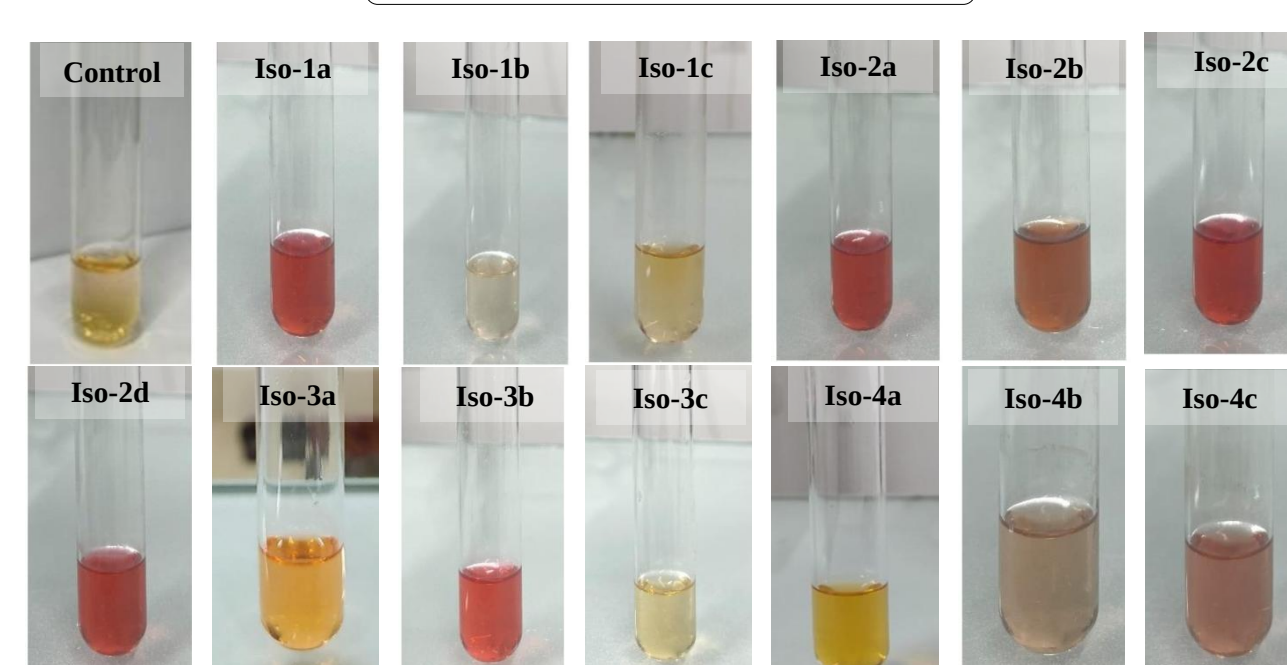
Zinc Solubilization Test



Siderophore Detection Assay

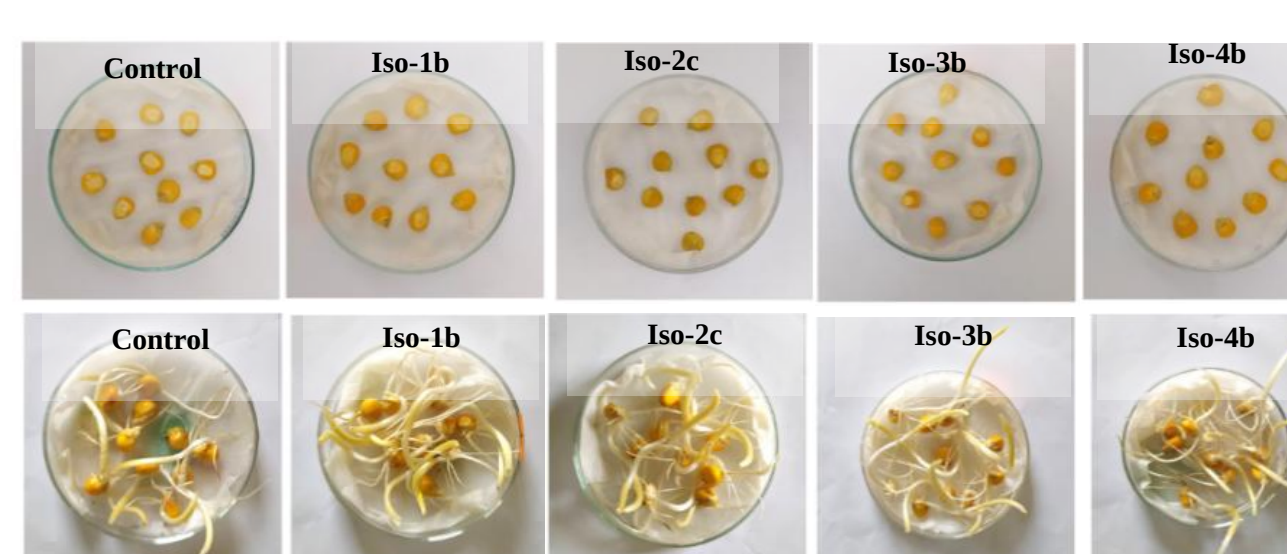


Ammonia Production Test



IAA Test

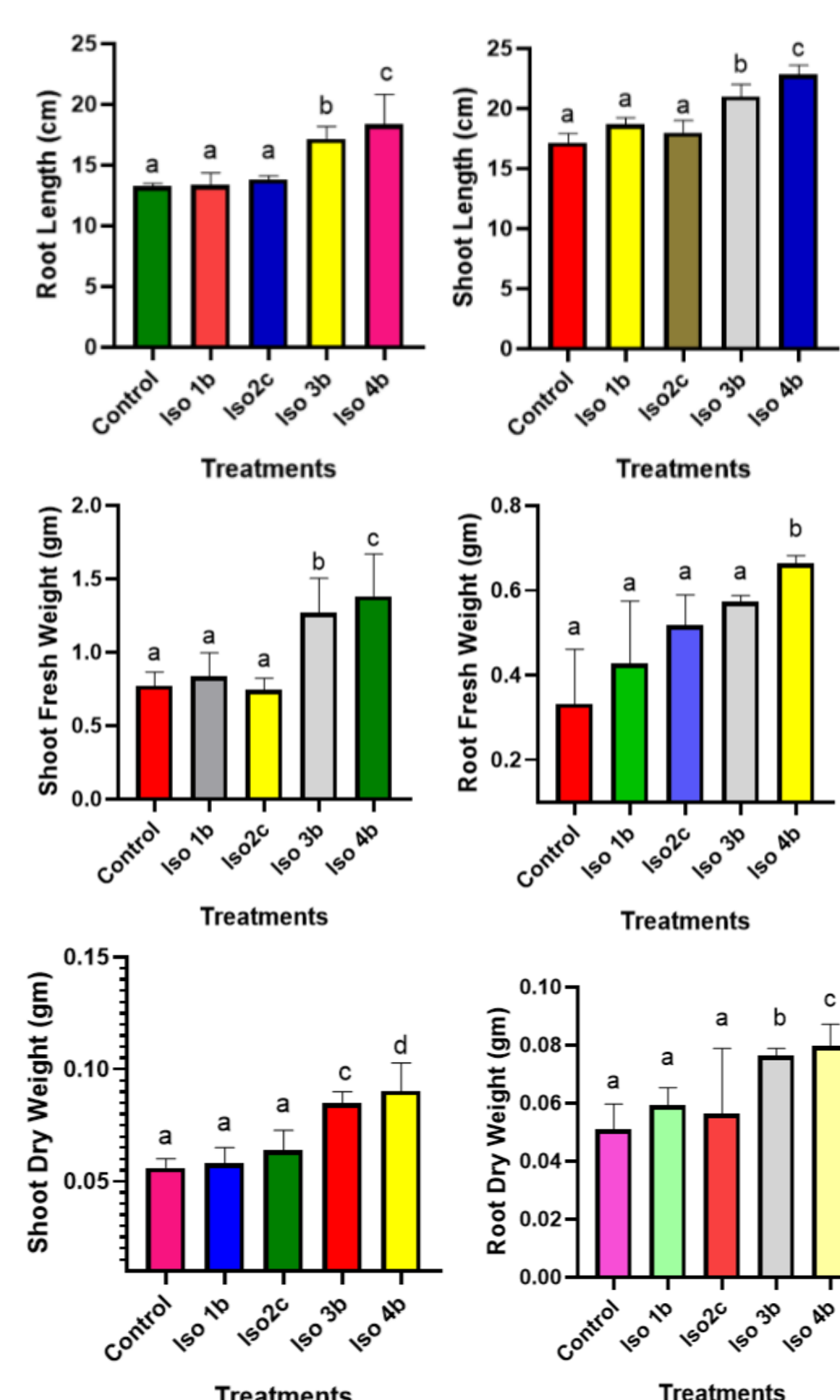
In Vivo Pot Experiments



Seed Germination Result

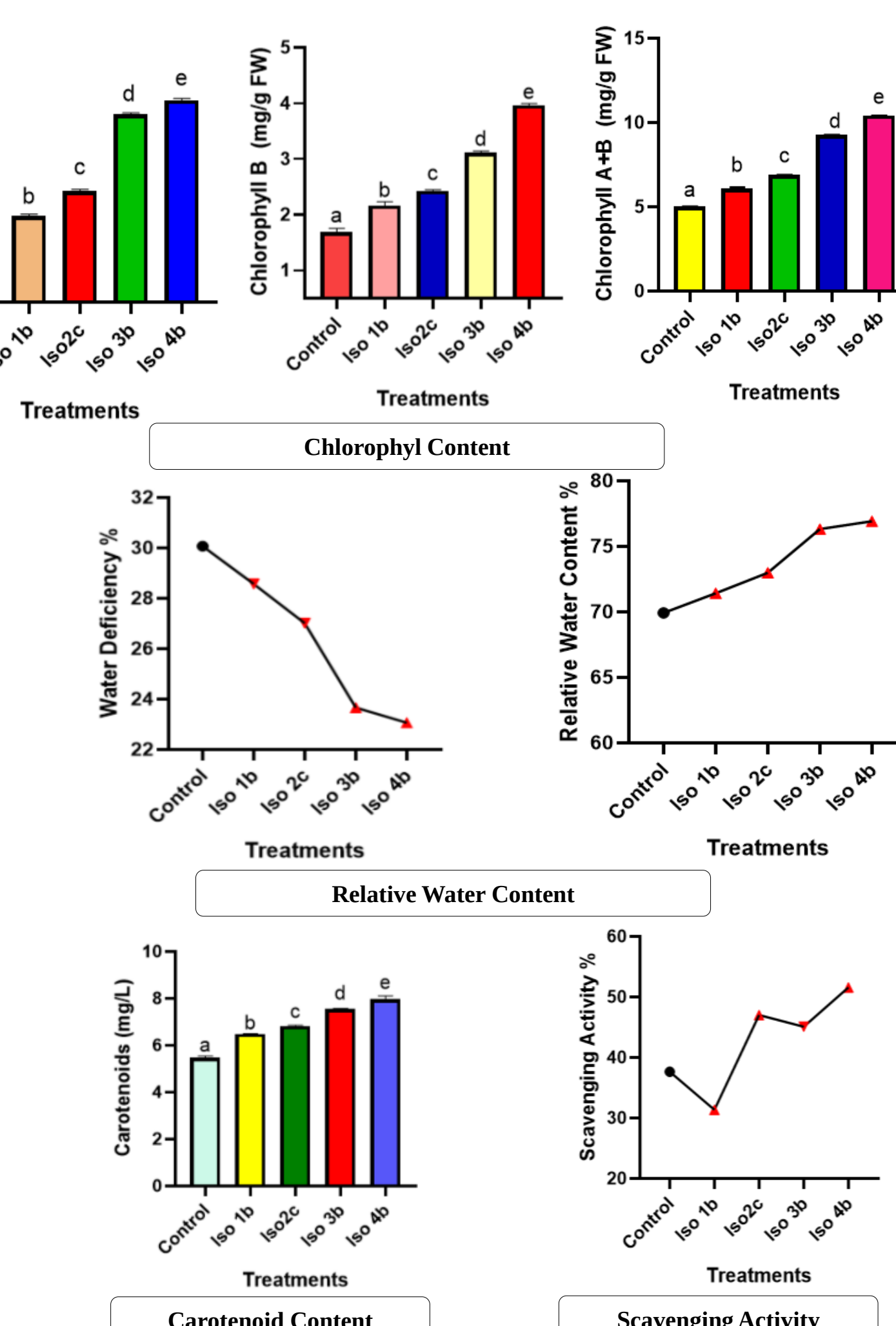


Summary of The Results of Morphological and Biochemical Test



Root and Shoot Fresh and Dry Weight (g)

Correlations Heatmap Among The Sixteen Plant Growth Parameters



CONCLUSION

In Rajshahi, Bangladesh, the study investigates the detrimental effects of chemical fertilizers on the production of maize, paying special attention to endophytic and rhizospheric bacteria. Four important bacterial isolates were found through a variety of biochemical tests to be able to support plant growth by means of processes like nitrogen fixation and the solubilization of essential nutrients. Chlorophyll content, root and shoot biomass, and maize seed germination were all markedly increased by treatments with these isolates. The study emphasizes the need for more research into the effects of beneficial bacteria while also highlighting their potential in agriculture.

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- Timsina J, Wolf J, Guilpart N, Van Bussel L, Grassini P, Van Wart J, et al. Can Bangladesh produce enough cereals to meet future demand? Agricultural systems. 2018;163:36-44.
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